Water Quality C - Water Quality - CWRU 2021 Invitational - 02-06-2021

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1. (1.00 pts) In a partially mixed estuary salt water is mixed _____ and freshwater is mixed _____.

- A) downward, upward
- B) upward, downward
- C) horizontally, vertically
- D) vertically, horizontally

2. (1.00 pts) What is the freezing point of water?

- A) 0 F
- B) 34 F
- C) -18 F
- D) 32 F

3. (1.00 pts) What features may vary along the length of an estuary? (Circle all that apply)

(Mark **ALL** correct answers)

- [x] A) Salinity
- [ ] B) Surface Tension
- [x] C) Light Levels
- [x] D) Temperature

4. (1.00 pts) True or False. Kelp forests occur in warm, nutrient rich areas of shallow coastal waters.

- True
- False

5. (1.00 pts) **Name two ways in which human activity is or has been a threat to estuaries**

Expected Answer: Coastal development, introduction of invasive species, over fishing, dams, and global climate change

6. (1.00 pts)
Your teacher brings you to a nearby lake for a class field trip and explains that due to low amounts of nitrogen, organisms in the lake must compete for resources. “What does your teacher say would be the outcome of adding a large amount of nitrogen to the lake?"

Expected Answer: If we added an abundance of a limiting resource, eutrophication would occur resulting in algal blooms, tainted drinking water, affecting the food chain.

7. (1.00 pts) Between point source pollutants and non-point source pollutants, which is more likely to enter an estuary?

Expected Answer: Non-point source

8. (1.00 pts) How does Carbon Dioxide regulate the pH of Seawater?

Expected Answer: The carbon dioxide-carbonic acid-bicarbonate system functions as a buffer to keep the pH of seawater in a narrow range.

9. (1.00 pts) What is the rate at which pressure generally increases?

- (●) A) 1 atm/10 m water
- ( ) B) 1 atm/15 m water
- ( ) C) 1 atm/8 m water
- ( ) D) 1 atm/5 m water

10. (1.00 pts) An oil spill has recently occurred in an area near you. Your area is known to be a popular tourist attraction for exotic birds. What can you expect relatively soon?

Expected Answer: Oil reduces the water repellency of bird’s feathers, exposing it to harsh conditions. Probably many bird deaths and a decrease in tourism.

11. (1.00 pts) How do mangroves offer protection from natural disasters?

Expected Answer: They offer protection from catastrophic events, such as tsunami, tropical cyclones and tidal bores and can dampen shoreline erosion

12. (1.00 pts) Please label the following with their respective survivorship curve type.

Squid

Turtle

Human

Seagull

Expected Answer: Squid (Type 3) Turtle (Type 3) Human (Type 1) Seagull (Type 2)

13. (1.00 pts) **Describe (to the best of your ability) a proper survivorship curve for a turtle.**

Expected Answer: Should start with a high y value at x = 0 and drop quickly to a small y value at around x = 2. Then the graph should approach the x axis but never touch it (asymtote-like)

14. (1.00 pts) What type of filtration method is the most common in municipal wastewater treatment? (Circle all that apply)

(Mark **ALL** correct answers)

- [x] A) Sand
- [ ] B) Fabric
- [ ] C) Bacterial
- [x] D) Gravel

15. (2.00 pts) Discuss two drawbacks of using mechanical filtration systems at a wastewater treatment plant.

Expected Answer: Debris will remain trapped in the filter and continue to be exposed to the water until it is manually removed by the aquarium keeper. Organic debris will continue to break down while in the filter and release ammonia Not a lot of variety exists when it comes to mechanical filtration options

16. (1.00 pts) XVI - Please classify each section of a river. (headwater, lower reaches, midreaches) (Use the Image Document Provided)

Expected Answer: Left to right Headwaters, lower reaches, mid reaches

17. (1.00 pts) What are some drawbacks of the River Continuum Concept?

Expected Answer: It describes a perfect and even model without taking into account changing riverine disturbances and irregularities. Disturbances such as congestion by dams c natural events such as shore flooding are not included in the model.

18. (2.00 pts)
You and your environmentalist friend are taking a walk in the neighborhood and pass a new mansion being built with a stunning lawn and a swimming pool. Your friend shakes his head. You ask him why he is disappointed. What do you think he says to you?

Expected Answer: The most concentrated sediment releases come from construction activities, including relatively minor home-building projects such as room additions and swimming pools. More points if they talk about the harm to the environment.

19. (2.00 pts) What is the difference between an r and k selected species? Give an example of each?

Expected Answer: r-selected babies grow rapidly, and tend to be found in less competitive, low quality environments. ... K-selected species produce offspring that each have a higher probability of survival to maturity.

20. (2.00 pts) What is the difference between an r and k selected species? Give an example of each?

Expected Answer: r-selected babies grow rapidly, and tend to be found in less competitive, low quality environments. ... K-selected species produce offspring that each have a higher probability of survival to maturity.

21. (0.10 pts) Tie Breaker: List and describe as many natural cycles as possible

Expected Answer: Carbon, Nitrogen, Oxygen, Water, Phosphorus For each natural cycle they list they get 0.01 points. For each one described as well they get another 0.01 points for a total of 0.02 points for each natural cycle

22. (1.00 pts) What is the difference between Hermatypic Corals and Ahermatypic Corals?

Expected Answer: Hermatypic Corals - Corals that form large colonies called reefs and have a symbiotic relationship with the dinoflagellate Zooxanthellae Ahermatypic Corals - Corals that are solitary or form small colonies- they often lack the symbiotic relationship with Zooxanthellae and do not help build reefs

23. (1.00 pts) What are some requirements for Coral Reef Formation? (Circle all that apply)

(Mark **ALL** correct answers)

- [x] A) Low nutrient waters
- [x] B) Solid base with a substance/layer for attachment
- [x] C) Decent water circulation
- [ ] D) High deposits of salt in the area
- [ ] E) All of the above

24. (1.00 pts) Please list the stages of coral reef development in order.

Expected Answer: SPAWNING >> FERTILIZING >> PLANULAE LARVAE SETTLEMENT >> CLONING.

25. (1.00 pts) How would you discern the age of a coral reef

Expected Answer: Count the number of coral growth layers

26. (1.00 pts) List 2 threats to coral reefs and how they affect the reef. (consequences)

Expected Answer: Threats: Chemical pollutants, Excess nutrients, Sedimentation, Coral bleaching, Coral diseases, Climate change and ocean acidification, overfishing. Consequences: loss of sensitive reef builders, reduced species richness, dominance of macroalgae, lowered coral reef resilience, loss of ecosystem good and services

27. (1.00 pts) VI - What coral disease is displayed? (Use provided image document)

Expected Answer: Yellow Band/Blotch Disease

28. (1.00 pts)
Coral bleaching has become a problem in many coral reefs around the globe. Please explain what causes coral bleaching as well as what happens during coral bleaching.

Expected Answer: Coral bleaching results when the symbiotic zooxanthellae (single-celled algae) are released from the original host coral due to stress (e.g., unusually high or low temperatures, high or low salinities, or excessive sedimentation). The coral loses its pigment (color) when the zooxanthellae are expelled. Bleaching can inhibit the ability of coral to recover from small-scale tissue damage; increase partial or total mortality; reduce reproduction success; or increase susceptibility to disease or other stresses.

29. (2.00 pts) How might Coral bleaching affect us humans? (two ways)

Expected Answer: Coral bleaching impacts peoples' livelihoods, food security, and safety. Coral reefs are natural barriers that absorb the force of waves and storm surges, keeping coastal communities safe. Without them, we must rely on manmade seawalls that are expensive, less effective, and environmentally damaging to construct. Bleached coral also compounds the overfishing crisis by removing links in the food web and depriving some fish and crustacean species of a place to spawn and develop. Tourism (another point if mentioned)

30. (1.00 pts) XI - Identify this organism that dwells in reefs and what it does to coral. (Use provided image document)

Expected Answer: Coral-feeding snails (Coralliophila abbreviate) remove tissue in a semicircular pattern.

31. (1.00 pts) Explain what Zooxanthellae give to the coral and what the coral do to the materials received by the Zooxanthellae.

Expected Answer: zooxanthellae supply the coral with glucose, glycerol, and amino acids, which are the products of photosynthesis. The coral uses these products to make proteins, fats, and carbohydrates, and produce calcium carbonate.

32. (1.00 pts) Please describe a simple coral reef food web (1 example in each category, predator, producer, consumer etc).

Expected Answer: As long as they provide one example of each it should be fine. Ex. Shark - Fish - Detritus - Coral

33. (1.00 pts) What do sponges do for a coral reef?

Expected Answer: Algae and corals in coral reefs produce organic material. This is filtered through sponges which convert this organic material into small particles which in turn ar absorbed by algae and corals

34. (1.00 pts) Which zone is arguably the most diverse zone of a coral reef.

- A) off reef floor
- B) reef lagoon
- C) the reef face
- D) the reef flat

35. (1.00 pts) How do Corals reproduce?

- A) Sexually
- B) Asexually
- C) Both

36. (2.50 pts)
Overfishing is a known threat to coral reefs. What regulations have been put in place to prevent or mitigate the effects of overfishing? (Up to 5 regulations worth 0.5 each)

Expected Answer: Establish or expand Marine Protected Areas (MPAs) and Locally Managed Marine Areas (LMMAs) establish sustainable fisheries policies and practices, such as limiting the type of fishing gear to reduce bycatch or establishing spatial or seasonal closures that reduce the catch of rare species, juveniles, and spawning aggregations. conducting law enforcement patrols, clarifying governance and building awareness among fishers promoting sustainable alternative livelihoods Increase stewardship for coral reefs through education and awareness-building

37. (1.00 pts) What is a common way to track ocean acidification?

Expected Answer: Aragonite Saturation

38. (0.01 pts) Tie Breaker: Chemical Equation for previous question

Expected Answer: $Ca^{2+} + CO_3^{2-} \leftrightarrow CaCO_3$

39. (1.00 pts) Describe the MPN method.

Expected Answer: The MPN (most probable number) method uses a test tube full of media with a smaller inverted test tube inside which captures carbon dioxide gas released from the growth of coliform bacteria. A series of dilutions and replicates are set up, and those producing gas in 24 hrs at 35 degrees C are counted. A statistical analysis is used to determine the most probable number of bacteria cells present.

40. (2.00 pts) IV - Is the above graph an accurate representation of depth distribution of dissolved oxygen, pH and CO_2? Why or why not? (Use provided image document)

Expected Answer: 1 point for yes 1 point for description Yes pH of the ocean normally ranges from 8.1 to 8.4 at the surface pH is directly influenced by carbon dioxide levels of Animal Activity: Increases acidity (lower pH values); respiration decreases O_2 levels and increases CO_2 levels o Plant Activity: Decreases acidity (higher pH values); increases O_2 and decreases CO_2 from the water through photosynthesis

41. (2.00 pts)
V - What information from the graph above may be of the greatest concern for scientists studying climate change, looking for definitive proof of increasing CO_2 in the ocean?(Use the image document provided)

Expected Answer: 1 point for a 1 point for b a - The largest decreases in aragonite saturation have occurred in tropical waters; however, decreases in cold areas may be of greater concern because colder waters typically have lower aragonite saturation levels to begin with. b - CO_2 dissolves more readily in cold water than in warm water

42. (2.00 pts)

VI - The graph above shows the average distance moved by approximately 105 marine species over a 40 year period along the Northeast coast and in the eastern Bering Sea. How may this data be indicative of global warming?

Expected Answer: Award partial using discretion Changes in water temperature can affect the environments where fish, shellfish, and other marine species live. Certain fish specie naturally migrate in response to seasonal temperature changes, moving northward or to deeper, cooler waters in the summer and migrating back during the winter. As climate change causes the oceans to become warmer year-round, populations of some species may adapt by shifting away from areas that have become too warm and toward areas that were previously cooler. Along U.S. coasts, this means a shift northward or to deeper waters that may have a more suitable temperature. As smaller prey species shift their habitats, larger predator species may follow them.

43. (1.00 pts) Nitrate levels above _____mg/L are, by EPA standards unsafe.

10

44. (1.00 pts) Turbidity is measured in _____ using a _____.

Expected Answer: 0.5 for each blank Turbidity is measured in NTU: Nephelometric Turbidity Units. The instrument used for measuring it is called nephelometer or turbidimeter

45. (1.00 pts) What is the accepted temperature range and time of incubation for the BOD test?

Expected Answer: 20 +/-1°C and 5 days

46. (4.00 pts) X - Determine the BOD of each seed dilution, then calculate the average seed BOD. (Use provided image document)

Expected Answer: Award points using discretion Bottle #1 BOD = [(7.95 - 5.20) x 300]/3 = (2.75 x 300)/3 = 275 Bottle #2 BOD = [(7.95 - 3.85) x 300]/6 = (4.10 x 300)/6 = 205 Bottle #3 BOD = [(7.90 - 2.40) x 300]/9 = (5.50 x 300)/9 = 183 Bottle #4 BOD = [(7.85 - 1.35) x 300]/12 = (6.50 x 300)/12 = 162 Average seed BOD = (275 + 205 + 183 + 162)/4 = 825/4 = 206 mg/L

47. (2.00 pts) What does the average seed BOD you determined in the above question tell you about the sample? (Continuation of previous problem)

Expected Answer: This value represents 206 mg/L BOD exerted by 300 mL of the seed material. In other words, a 300 mL sample of the undiluted seed material would use 206 mg/L DO if incubated at 20°C for five days (assuming that oxygen was available to the sample).

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